ARTIFICIAL INTELLIGENCE

HAND GESTURE RECOGNITION USING BAG-OF-VISUAL-WORDS

Prepared by:

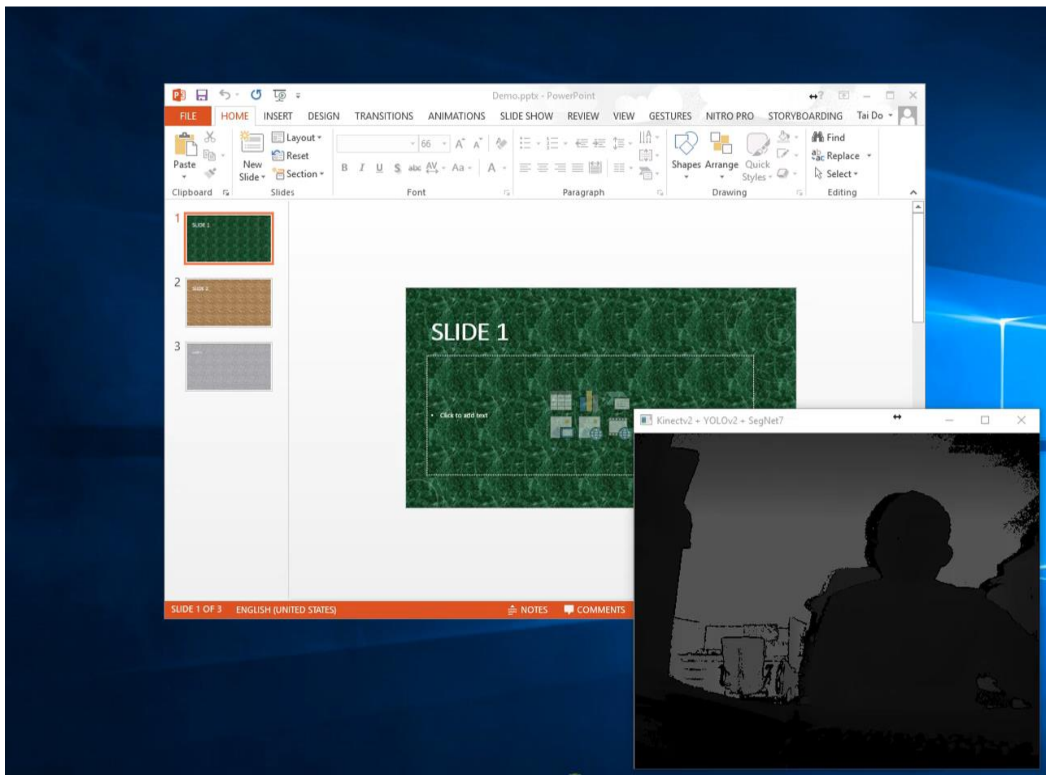
Sutdent Name (ID)

Date

Agenda

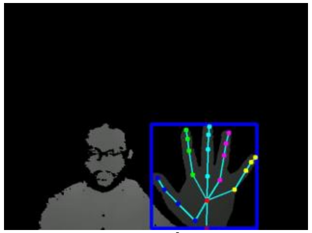
1. Introduction

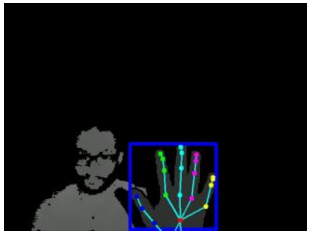
- **Dynamic Hand Gesture Recognition:**
  - exploiting more the temporal features from the handshape sequence
  - challenging due to its small size, complexity, and self-occlusion
  - applying in human-machine interaction

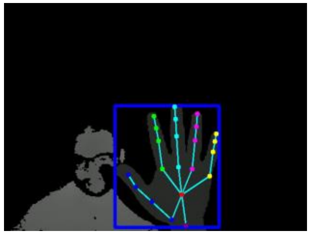


1. Introduction

- **Our problem:**
  - Dynamic Hand Gesture Recognition on DHG Dataset
  - Hand shape features based on Skeleton information and Depth image


Grab


Tap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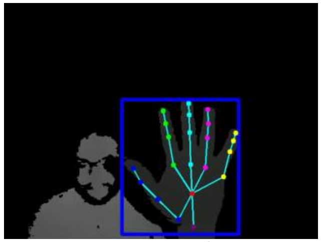
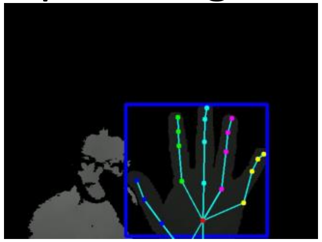
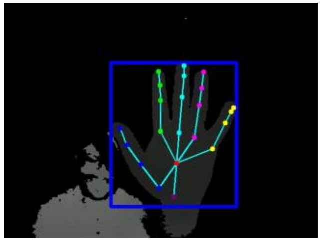
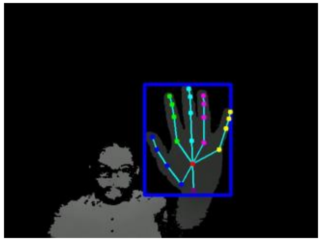

Pinch


Rotation CW


Rotation CCW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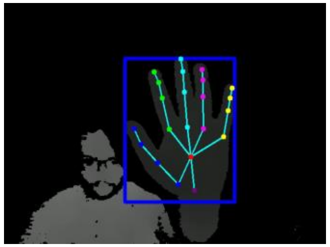
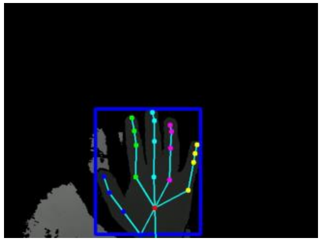
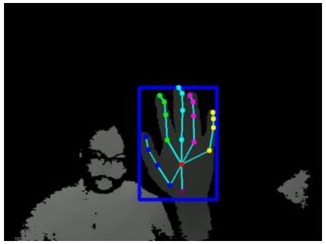
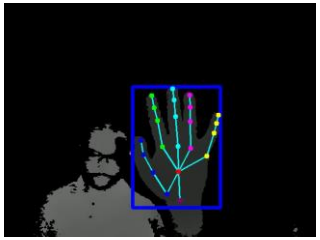
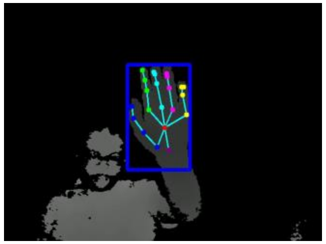
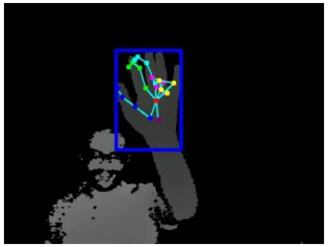

Swipe Left

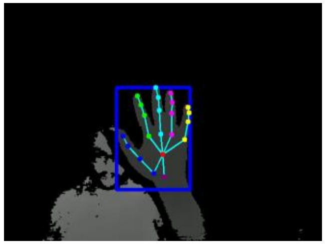

Swipe Up


Swipe Down


Swipe X


Swipe V

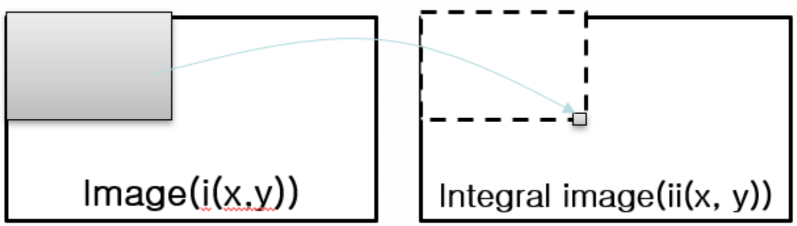

Swipe +

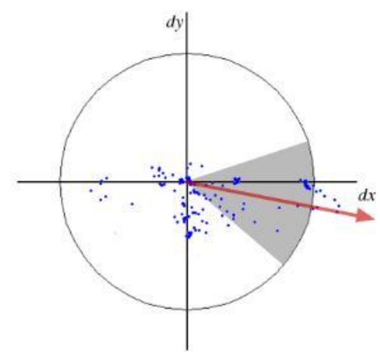

Shake

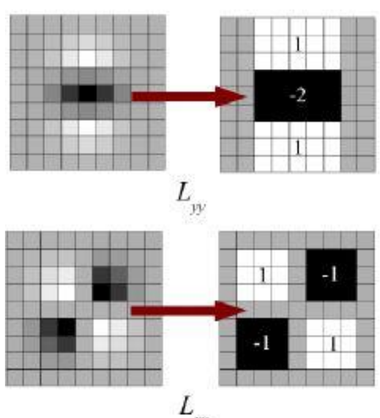
[1] Q. De Smedt, H. Wannous, and J.-P. Vandeborre, "***Skeleton-based dynamic hand gesture recognition***," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, 2016, pp. 1–9.

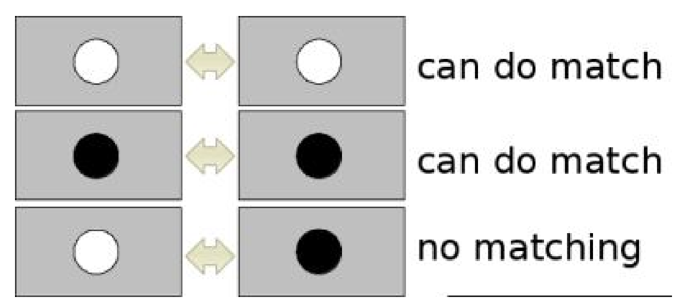
2. Related Research and Key Technology

- **Speeded Up Robust Features (SURF):**
  - approximates Laplacian of Gaussian with Box Filter using Integral image
  - feature description uses wavelet responses
  - sign of Laplacian for underlying interest point





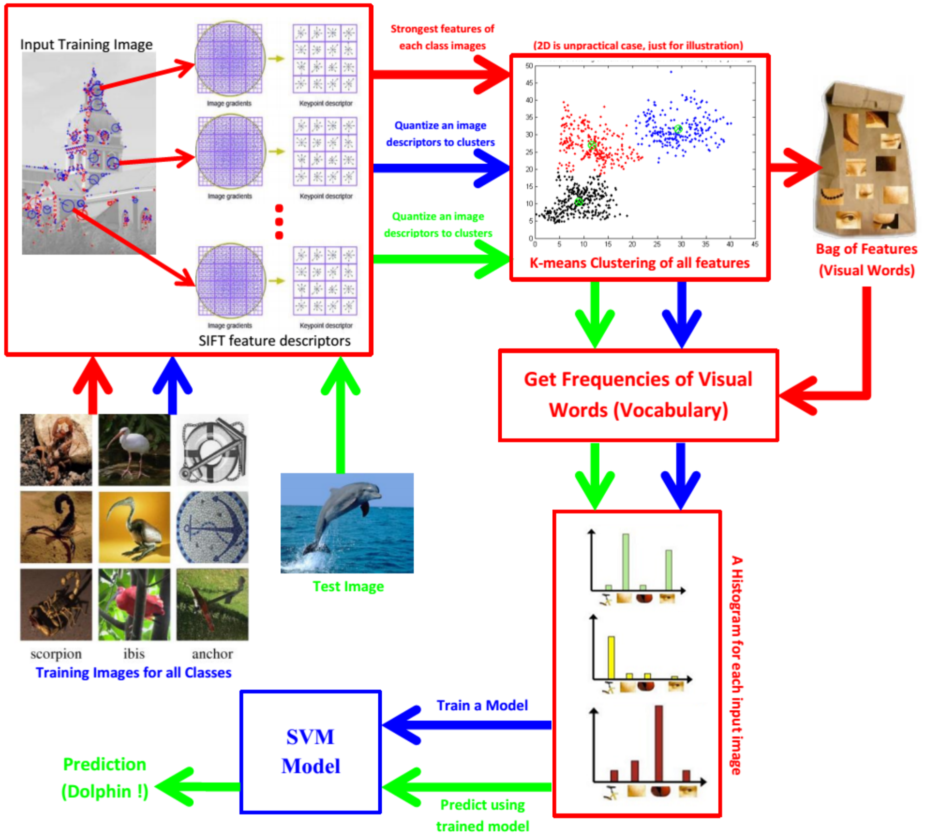




[1] H. Bay, T. Tuytelaars, and L. Van Gool, "***Surf: Speeded up robust features***," in *European conference on computer vision*, 2006, pp. 404–417.

2. Related Research and Key Technology

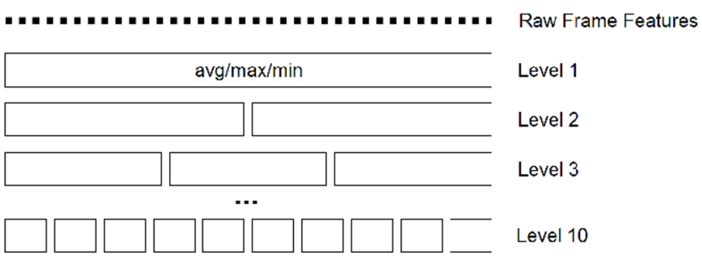
- **Bag-Of-Visual Words:**
  - An image is represented by a collection of "visual words" and their corresponding counts given a universal dictionary.



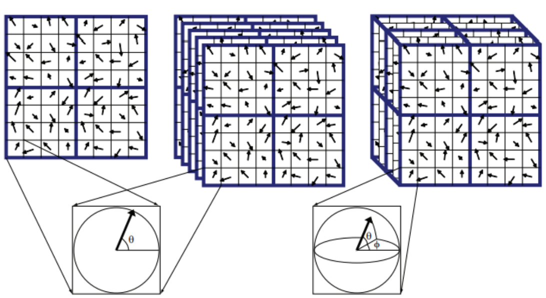
[1] Fei-Fei, Li, and Pietro Perona. "***A bayesian hierarchical model for learning natural scene categories***." *2005 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR'05)*. Vol. 2. IEEE, 2005.

2. Related Research and Key Technology

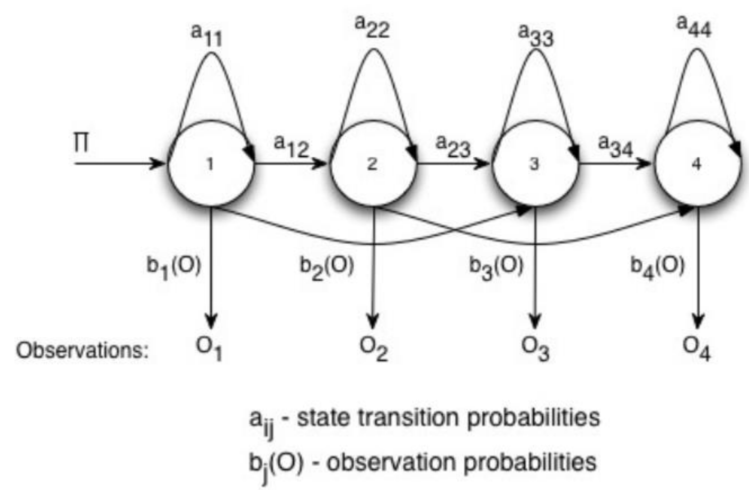
- **Temporal Model:**
  - descriptors which carry spatial and temporal information
  - modeling sequences of spatial descriptors via temporal classiers



Temporal Pyramid Model[1]



3D SIFT[2]

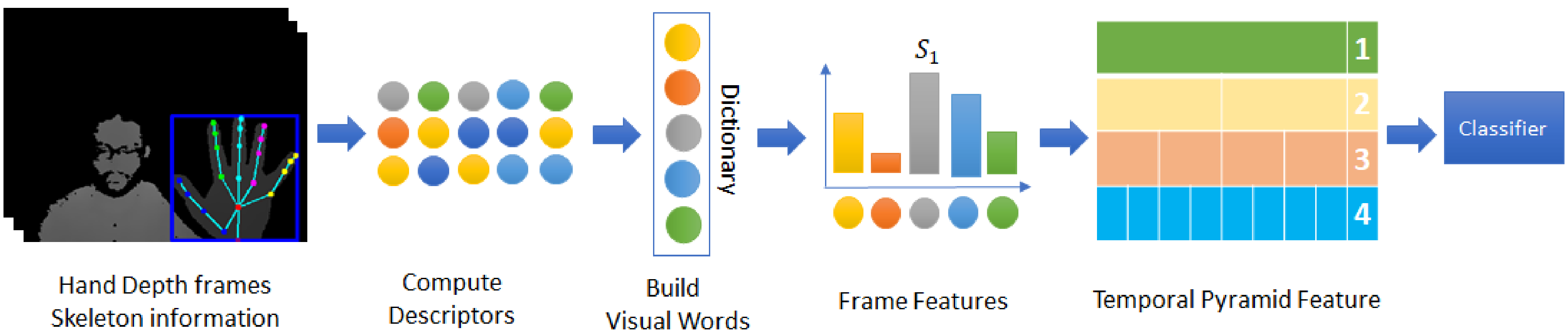


Hidden Markov Model

[1] Choi, Jaesik, Won J. Jeon, and Sang-Chul Lee. "Spatio-temporal pyramid matching for sports videos." Proceedings of the 1st ACM international conference on Multimedia information retrieval. ACM, 2008
[2] Scovanner, Paul, Saad Ali, and Mubarak Shah. "***A 3-dimensional sift descriptor and its application to action recognition***." *Proceedings of the 15th ACM international conference on Multimedia*. ACM, 2007.

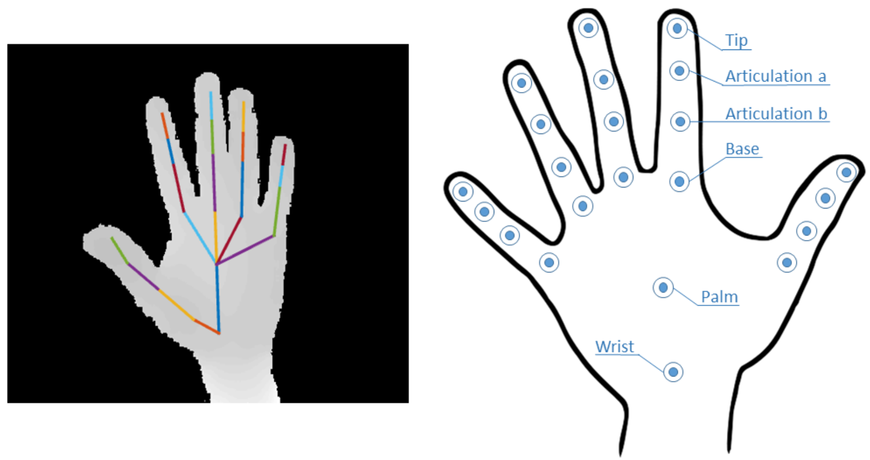
3. Project Contents and Algorithm

- **Overview:**
  - (1) Extracting hand shape feature: skeleton information, depth image
  - (2) Clustering local patches using k-means and Gaussian Mixture Model
  - (3) Computing frame feature from visual-words
  - (4) Extracting temporal pyramid feature
  - (5) Classifying the gesture



3. Project Contents and Algorithm

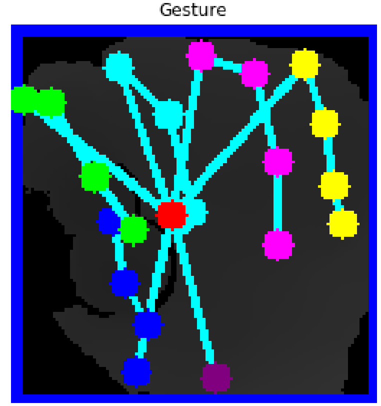
- **Hand shape feature from skeleton and depth:**



DHG Dataset annotation

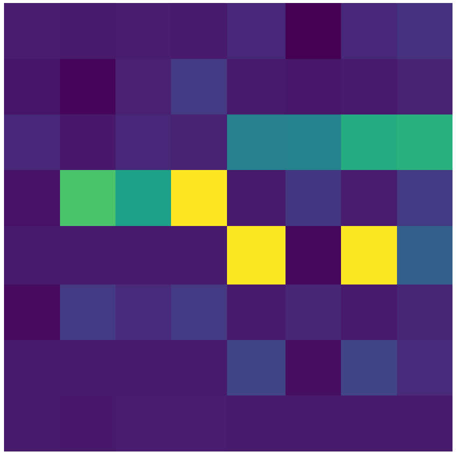
Depth Image

Gesture



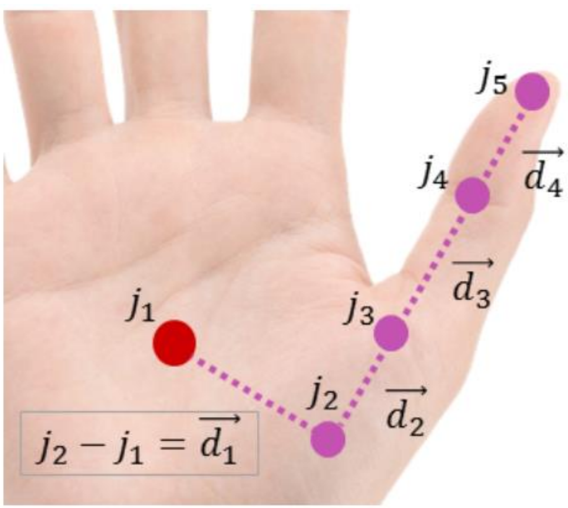
3. Project Contents and Algorithm

- **Depth image descriptor:**
  - Extracting keypoints and descriptors using SURF
  - 64 dimension feature vector

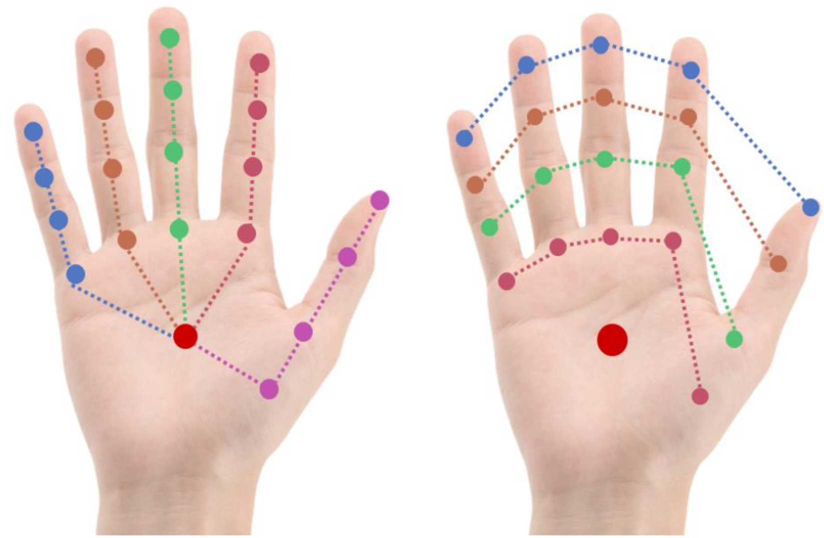


3. Project Contents and Algorithm

- **Skeleton information descriptor:**
  - Generate join pairs from 22 joints → 225 pairs
  - Calculate distance of a pair → 450-dimension feature vector



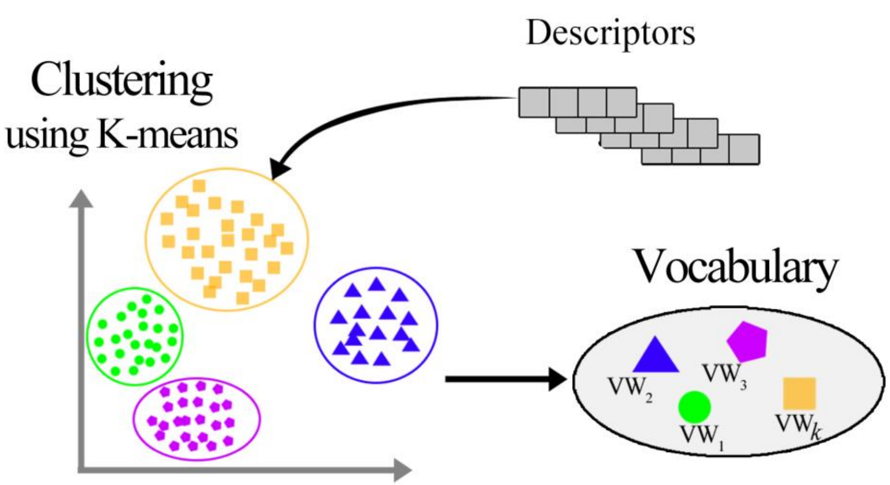
joint tuple distances



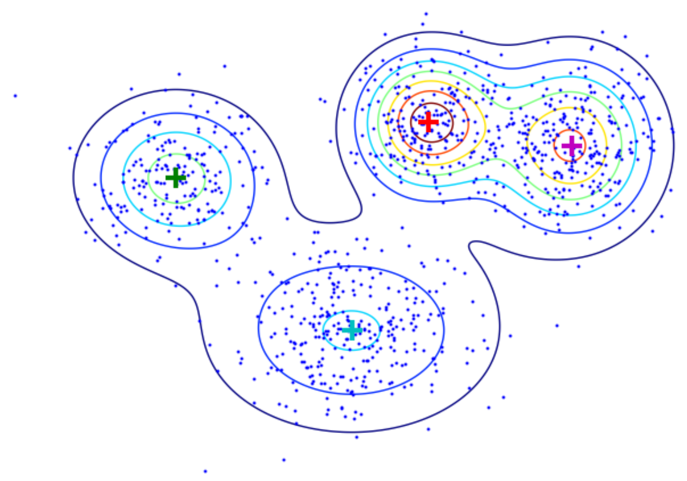
nine tuples sample for hand shape structure

3. Project Contents and Algorithm

- **Clustering with Gaussian Mixture Model and k-means:**
  - Choose k = 500 visual words as number of clusters
  - It takes 4 hours for clustering
  - With GMM, only use sub-set of data for clustering



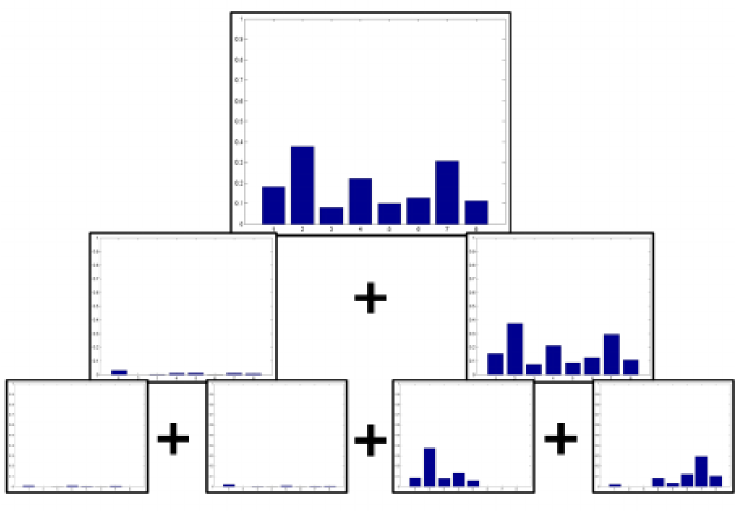
Bag-of-visual words with k-means



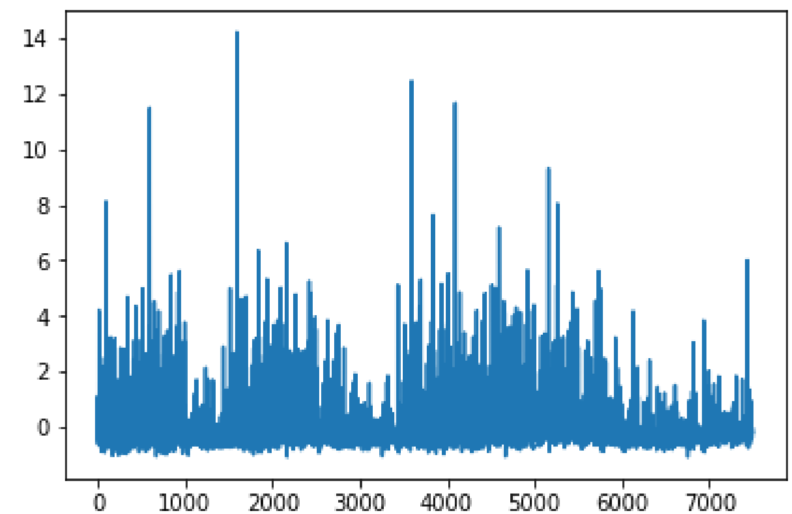
Gaussian Mixture Model

3. Project Contents and Algorithm

- **Temporal Pyramid Feature Extraction:**
  - Choose n = 4 level sub-sequences
  - With 500 visual-words, extracting 7500-dimension feature vector
- **SVM Classification:**
  - C = 100.0, gama = “auto”, probability = “True”, kernel = “rbf”



3 level temporal pyramid feature



An example of sequence feature

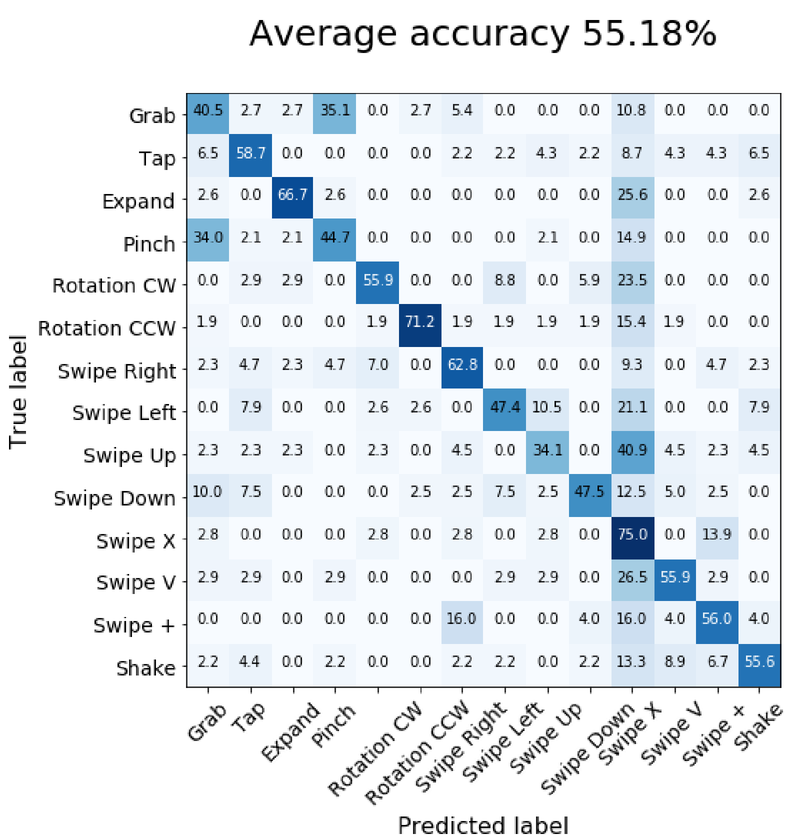
4. Experiment and Result

- **Proposed methods:**

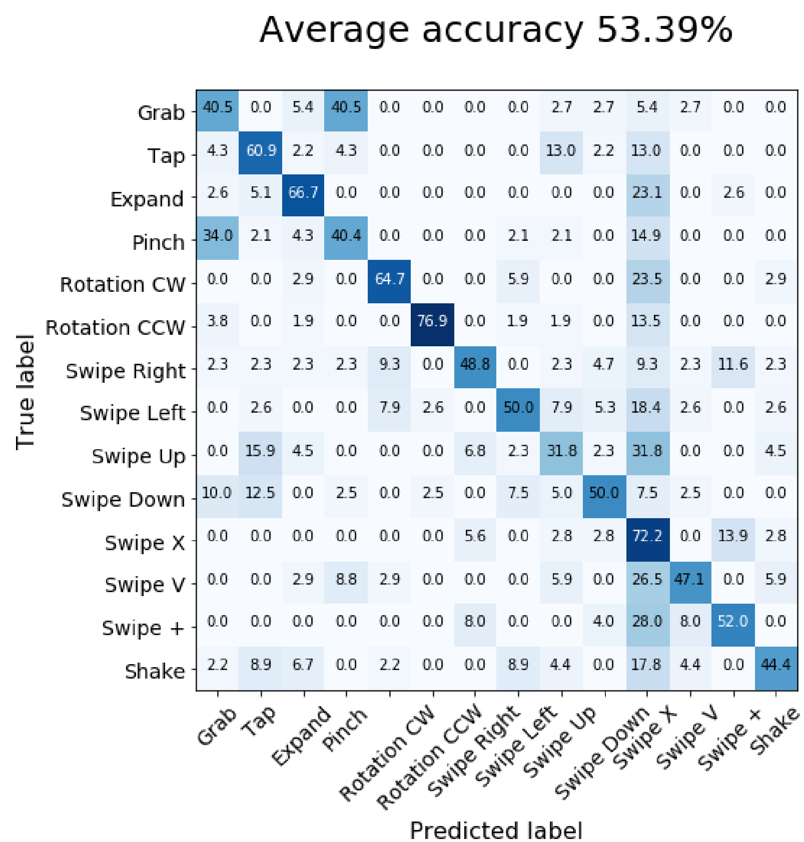
Method	Feature	Clustering	Operator
1	Skeleton	K-means	
2	Skeleton	GMM	
3	Depth	K-means	
4	Depth	GMM	
5	Fusion		Max
6	Fusion		Average
7	Fusion		Weighted

4. Experiment and Result

- **Skeleton Features:**



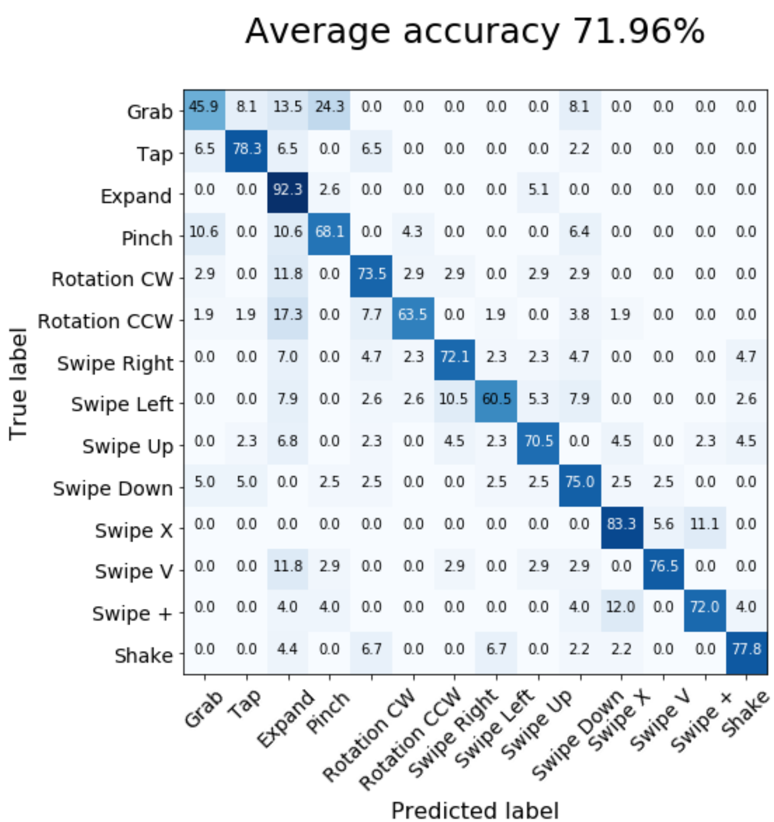
Gaussian Mixture Model



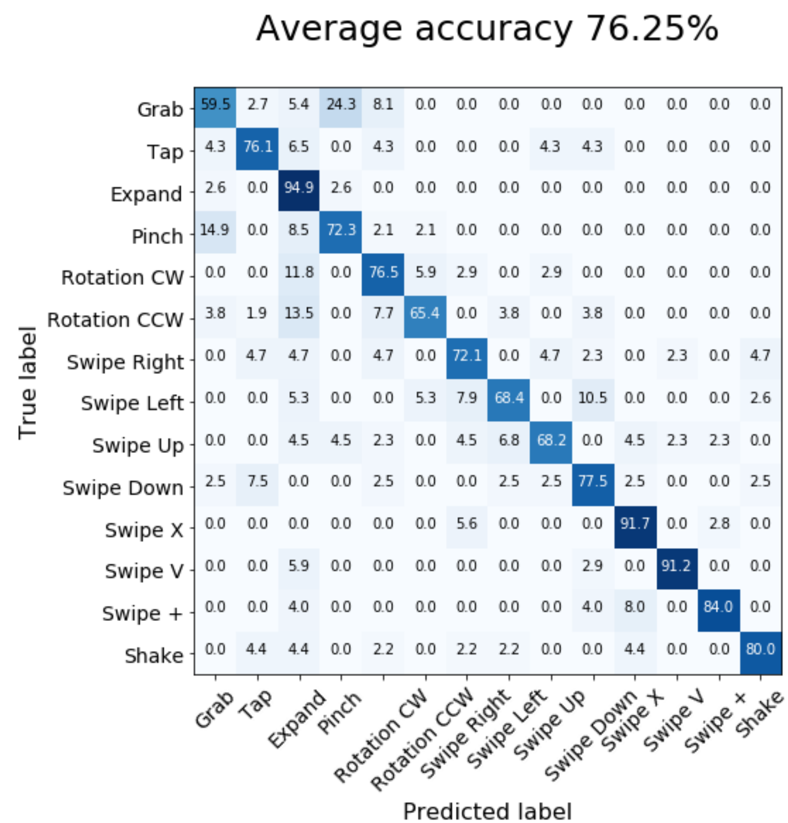
K-means

4. Experiment and Result

- **Depth Features:**



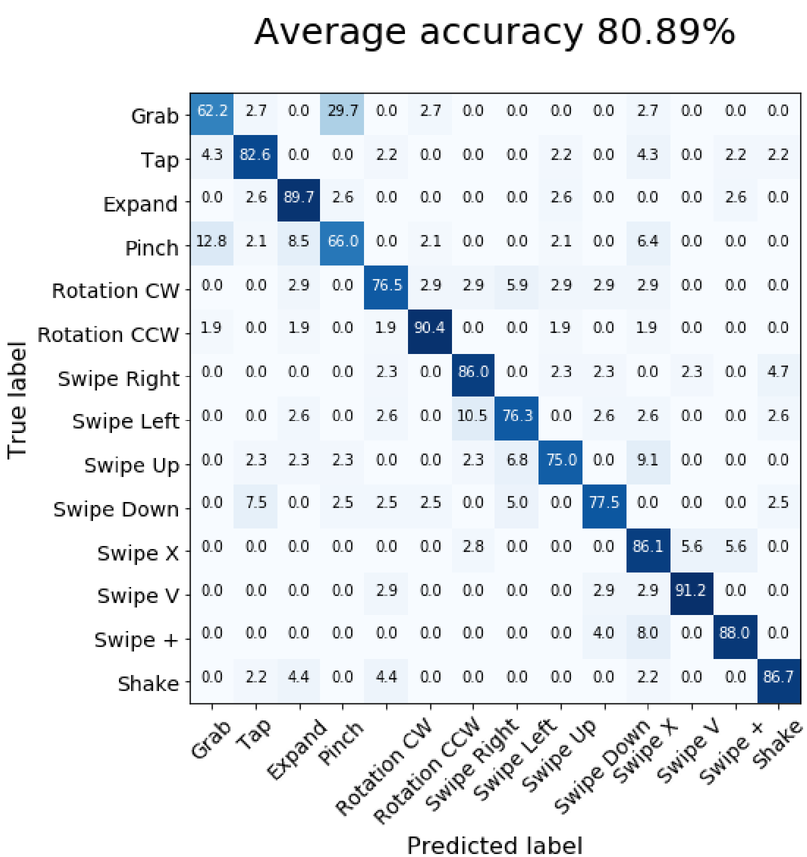
Gaussian Mixture Model



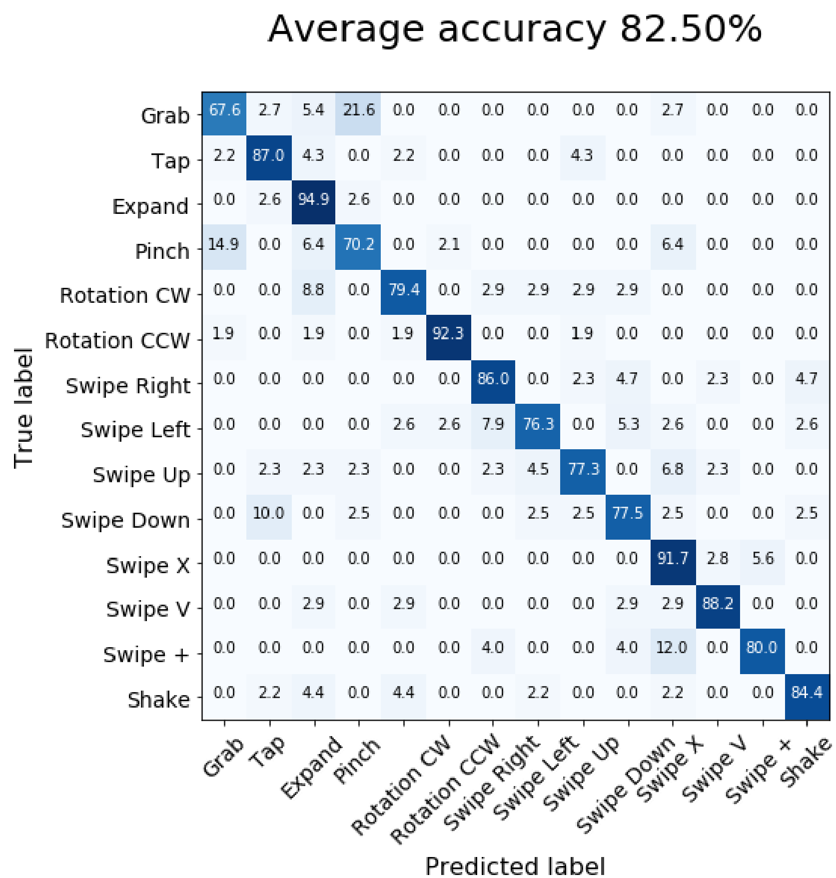
K-means

4. Experiment and Result

- **Max and Average Fusion:**



Max Fusion

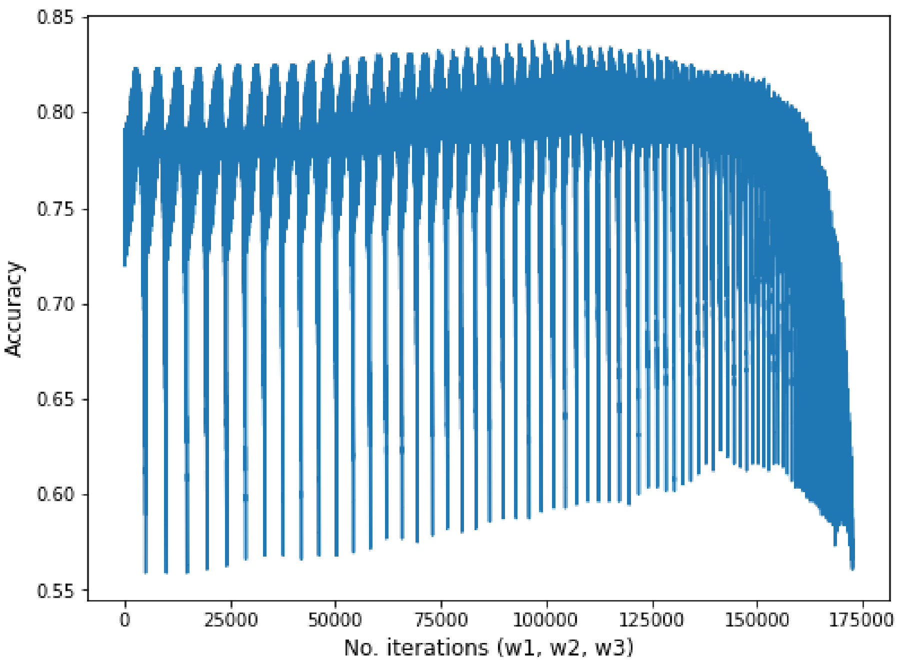


Average Fusion

4. Experiment and Result

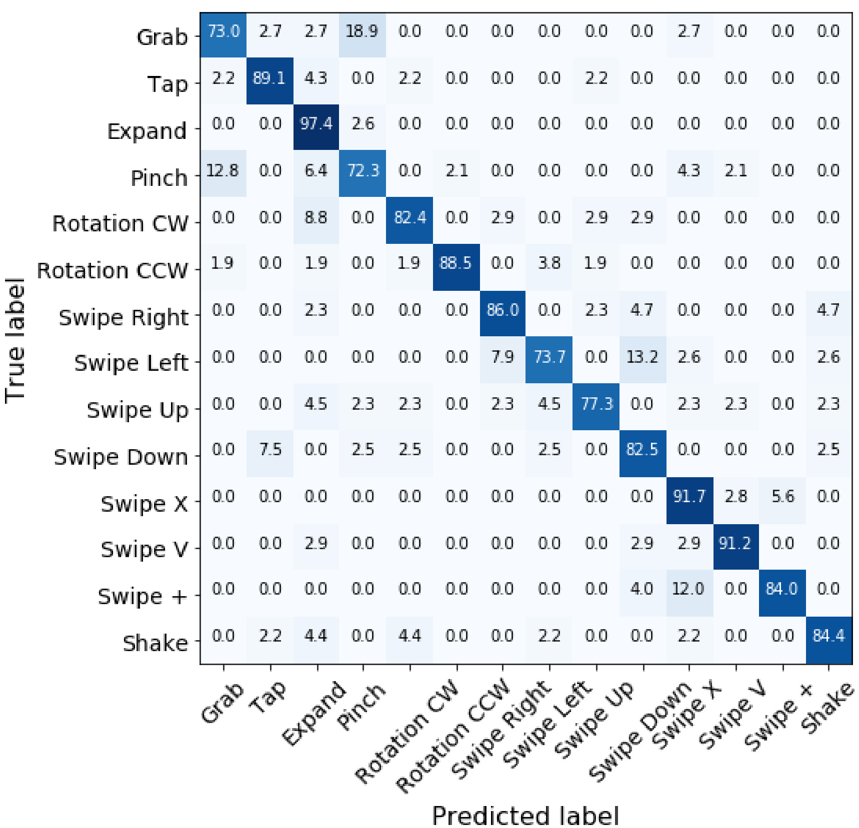
- **Weighted Fusion:**

$$w_1.model_1 + w_2.model_2 + w_3.model_3 + (1-w_1-w_2-w_3).model_4$$



Search Params

Average accuracy 83.75%



Weighted Fusion

4. Experiment and Result

- **Summary:**

Method	Feature	Clustering	Operator	Accuracy
1	Skeleton	K-means		53.39%
2	Skeleton	GMM		55.18%
3	Depth	K-means		76.25%
4	Depth	GMM		71.96%
5	Fusion		Max	80.89%
6	Fusion		Average	82.50%
7	Fusion		Weighted	83.75%

5. Conclusion

- Propose dynamic hand gesture recognition
- Base on hand shape feature with skeleton and depth information
- Use bag-of-visual-words with k-means and gaussian mixture model
- Extract temporal pyramid feature for sequence
- Recognition gesture with SVM Classification
- Ensemble results with maximum, average and weighted operator
- Achieve good accuracy on DHG Dataset

THANKS FOR LISTENING!